Viva Teaching Guide

Explainable Speech-Based Depression Detection

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Deployed model: **Random Forest** · Held-out: **75.0% Acc**, **63.6% recall** (n=32)
Framing: decision support — **not** diagnosis

Study in order. Each topic: meaning → your project → likely viva Q&A.

PART A — Basics (start here)

A1. What is depression (in this project)?

What it means

Depression is a mental health condition. Clinicians often use questionnaires like **PHQ-8 / PHQ-9**. A common cut-off is PHQ $\geq 10 \rightarrow$ depressed (binary).

Your project

You do **binary classification**:

- **Depressed** (PHQ binary = 1)
- **Non-Depressed** (PHQ binary = 0)

If Depressed, you also show a **research subtype/profile** (heuristic — not a medical diagnosis).

Viva Q: Is your system a doctor?

A: No. It is a **decision-support research tool**. A clinician must interpret it with other information.

A2. Why speech?

What it means

Depressed speech often shows:

- lower energy (quieter)
- flatter pitch (less variation)
- slower rate
- longer pauses

These are **acoustic / paralinguistic** cues (how you sound), not the words alone.

Your project

Speech-only deployed path (no video; text only used to cut interviewer speech from transcripts).

Viva Q: Why not only text?

A: Speech acoustics can work even when transcripts are limited; privacy-friendly; matches your deployed acoustic model.

A3. What is machine learning here?

What it means

1. Extract numbers from audio (**features**)
2. Train a model on labelled examples
3. Predict new audio
4. Measure how often it is correct (**metrics**)

Your project pipeline

```
Audio → participant speech → segments → 23 acoustic features
→ aggregate to one person vector → Random Forest → Depressed/Non-Depressed
→ explanations (occlusion, timeline, features) → Web UI
```

A4. Dataset basics (DAIC-WOZ / AVEC)

What it means

- **DAIC-WOZ:** clinical interview corpus (participant + virtual interviewer Ellie)
- **AVEC 2017:** challenge with **official PHQ labels** and train/dev/test splits

Your usable data

Set	n	Depressed	Non-depressed	Use
Train+Dev	95	37	58	Train + choose threshold
Held-out test	32	11	21	Final score only
Total usable	**127**	48	79	—

Critical point

Folder names (`depressed/`, `non depressed/`) are **not always correct**. You use **official PHQ labels** (~39 folder mismatches corrected).

Viva Q: Why not use folder labels?

A: They are noisy. Official PHQ is the clinical ground truth for AVEC.

PART B — Your method (core story)

B1. Preprocessing

Steps

1. Load audio at **16 kHz**, mono
2. Prefer **participant-only speech** using transcript (remove Ellie)
3. Segment into **5 s** windows, **50% overlap**
4. Cap duration / number of segments for consistency

Why participant-only?

Training used patient speech only. If you upload full interview (Ellie + patient), the model often wrongly says Non-Depressed.

Demo tip: Upload `321_AUDIO.wav` or `325_AUDIO.wav` (filename must contain the ID).

B2. Features (23 acoustics)

What it means

Handcrafted numbers humans can interpret, e.g.:

- pitch mean / std
- energy
- speech rate
- pause ratio
- spectral features
- MFCCs

Aggregation

Many segments → one participant vector using:

mean, std, median, p25, p75

Viva Q: Why aggregate to participant level?

A: Final clinical label is per person. Segment-level scoring can leak speakers and inflate accuracy.

B3. Model — Random Forest (deployed)

What it means

Random Forest = many decision trees; majority / probability vote. Good for tabular features and smaller clinical datasets.

Your settings

- `n_estimators=600`
- `max_depth=8`
- `min_samples_leaf=1`
- `max_features=sqrt`
- `class_weight=balanced_subsample`
- `random_state=42`
- Threshold ≈ **0.49** (chosen on train+dev only, **never** on test)

Dual pathway

- **Deployed:** Random Forest + occlusion
- **Baseline only:** CNN + Grad-CAM

Viva Q: Why not CNN as final?

A: CNN was an early prototype. After official-label comparison, RF had a better deployable accuracy–recall balance. CNN remains for Grad-CAM demo only.

Viva Q: Why Extra Trees in your thesis history?

A: Extra Trees was a previous candidate (~59.4% Acc). Final deployed model is **Random Forest (75.0%)**.

B4. Explainability (XAI)

Method	Used for	Meaning
Occlusion	Deployed RF	Remove one segment; if probability changes a lot → important
Timeline cards	Both	Time ranges + plain-language cues
Feature ranking	Acoustic	Which features matter
Spectrogram	Visual	Show the recording
Grad-CAM	CNN baseline only	Heatmap on spectrogram

Viva Q: Does explanation change the prediction?

A: No. XAI is **post-hoc**. The RF decides first; explanations justify it.

Viva Q: Is subtype diagnostic?

A: No. Heuristic research profile when prediction = Depressed.

B5. Web system

- **FastAPI** backend (`server.py`)
- Frontend upload + tabs

- Patient save / delete records
- Disclaimer: not a diagnosis

Tabs: Why · Type · Spectrogram · Timeline · Attribution · Features

PART C — Evaluation (memorise this table)

C1. Metrics (what each means)

Metric	Meaning (simple)
Accuracy	Overall % correct
Balanced accuracy	Average of recall & specificity (fairer if imbalanced)
Precision	Of predicted depressed, how many truly depressed
Recall / Sensitivity	Of truly depressed, how many you caught
Specificity	Of truly non-depressed, how many correctly non-depressed
F1	Balance of precision & recall
ROC-AUC	Ranking quality across thresholds
Confusion matrix	TN / FP / FN / TP counts

C2. Your held-out numbers (FINAL)

Random Forest, n = 32

Metric	Value
Accuracy	**75.0%**
Balanced accuracy	**72.3%**
Precision	**63.6%**
Depression recall	**63.6%**
Specificity	**81.0%**
F1	**63.6%**
ROC-AUC	**0.654**
Threshold	≈ **0.49**
Acc. 95% CI	**59.4% – 87.5%**

CM: TN=17, FP=4, FN=4, TP=7

Caveat: accuracy is **seed-sensitive**; always mention the CI.

C3. Why not higher?

- Small test set (32)
- Speech-only
- Clinical data is hard / noisy
- Honest official-label protocol (no cheating with leakage)

C4. Candidate comparison (one line each)

- **RF deployed:** best usable Acc + recall

- **Extra Trees:** older, 59.4% Acc
- **PHQ regression:** 68.8% Acc but only 45.5% recall → bad for screening
- **LoRA WavLM:** high recall, low accuracy → not deployed

PART D — Design / Implementation talking points

D1. One-minute architecture speech

> "Offline we train on official AVEC labels: extract participant speech, acoustic features, aggregate, choose Random Forest and threshold on train+dev. Online, the user uploads audio; FastAPI runs the same features through RF; occlusion and timelines explain the result; the UI shows Depressed/Non-Depressed with confidence and a disclaimer."

D2. Key files

File	Role
`train_official_acoustic.py`	Train deployed model
`depression_acoustic_candidate.pkl`	Saved RF + threshold
`predict.py`	Inference
`explain.py`	Occlusion / charts / text
`server.py`	API
`frontend/`	UI

D3. Commands

```
python3 train_official_acoustic.py
python3 -m uvicorn server:app --host 127.0.0.1 --port 8765
```

PART E — Research questions & contributions

E1. RQs

1. Can speech support honest binary detection under official labels? → **Yes, moderately (75% Acc, 63.6% recall)**
2. Can multi-level explanations make predictions inspectable? → **Yes (occlusion, timeline, features)**
3. Is XAI post-hoc? → **Yes**

E2. Contributions (list 5)

1. Official-label, leakage-aware pipeline
2. Deployed RF acoustic classifier
3. Multi-level explainability
4. Web decision-support app + patient records
5. Honest evaluation + candidate comparison

E3. Limitations (say these yourself — examiners like honesty)

1. Test n=32; wide CI
2. Seed-sensitive accuracy
3. Speech-only
4. Single corpus
5. No clinician user study
6. Subtype is heuristic

E4. Future work

Larger/multi-corpus data · clinician studies · fairness by demographics · better calibration · human-in-the-loop screening

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PART F — Demo script (viva practical)

1. Open http://127.0.0.1:8765
2. Enter patient details
3. Upload `321_AUDIO.wav` or `325_AUDIO.wav`
4. Analyze → expect **Depressed**
5. Show Why / Timeline / Features
6. Upload `303_AUDIO.wav` → expect **Non-Depressed**
7. End: "Decision support, not diagnosis. Held-out 75% Acc, 63.6% recall."

Avoid for depressed demo: 308, 309 (model misses them)

PART G — Rapid-fire Q&A (memorise)

Q: Input / output?

A: Voice recording → Depressed/Non-Depressed + confidence + explanations (+ subtype if depressed).

Q: Features?

A: 23 acoustics aggregated with mean/std/median/percentiles.

Q: Model?

A: Random Forest, threshold ≈ 0.49.

Q: Dataset size?

A: 127 usable; 95 train+dev; 32 test.

Q: Accuracy?

A: 75.0% held-out; recall 63.6%; CI 59.4–87.5%.

Q: Explainability?

A: Occlusion for RF; Grad-CAM only for CNN baseline.

Q: Why 75% not 95%?

A: Honest clinical speech task, small test set, speech-only, no label leakage.

Q: Can it diagnose?

A: No.

Q: What if full interview uploaded without transcript match?

A: Interviewer speech can bias toward Non-Depressed; system tries to use transcript when ID/filename matches DAIC data.

Q: Overfitting?

A: Speaker-independent official split; threshold not tuned on test; report CI.

PART H — Words you must NOT say

- “This diagnoses depression”
- “100% accurate”
- “Final dataset is 9 participants / 108 segments”
- “Final model is CNN with 75% F1 0.857” (that was prototype only)
- “Deployed model is Extra Trees” (old; now RF)

PART I — 30-second closing speech

> “I built an explainable speech system that classifies Depressed versus Non-Depressed using official AVEC labels, a Random Forest on acoustic features, and occlusion-based explanations in a web app. On 32 held-out participants it reaches 75% accuracy and 63.6% recall. It is intended as decision support, not diagnosis, and future work needs larger clinical validation.”

Study plan (2–3 days)

Day	Focus
Day 1	Parts A–B (basics + method) + say metrics aloud 10 times
Day 2	Parts C–E (evaluation, RQs, limits) + draw architecture from memory
Day 3	Part F demo practice + Part G rapid-fire with a friend

Aligned to your deployed Random Forest system. Prefer this over older thesis drafts that still say Extra Trees 59.4% as final.